Mechanobiological Regulation of Thymic Selection

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A computational model and scoping review comprising project with the fundamental question:
How do the mechanical properties of the thymic stroma influence thymic selection outcomes?

Computational Model: Investigating mechanobiological mechanisms of thymic selection

Central hypothesis. Thymic stromal stiffness influences selection through two coupled routes: direct modulation of TCR signaling (force-dependent bond kinetics, cytoskeletal priming, mechanotransductional state) and indirect modulation of thymocyte–stroma contact through stiffness-dependent migration.

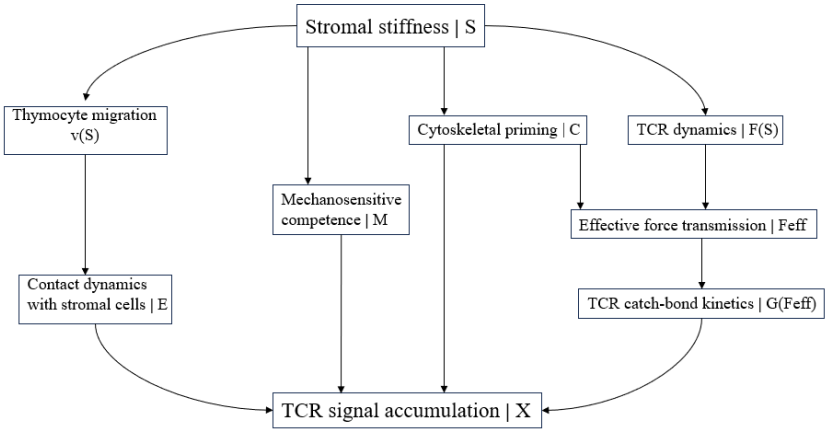


Figure-1: Mechanistic description of the model

Model in brief

- Hybrid agent-based/ODE model (designed using Mesa 3.x). This is a proof-of-concept model designed for hypothesis generation and should not be considered a well-developed digital twin.
- Model mechanism: Stiffness regulates biphasic migration, TCR dynamics (clustering, for eg.), cytoskeletal priming and mechanosensitive competence. C and F dictate how much force is transmitted to intracellular networks which modulate catch-bond kinetics (by application of force on TCRs). G(Feff) regulate how much TCR signal is produced, which requires stromal cell contact and is amplified by C and M.

Main results

Stiffness–selection relationship is non-monotonic. Positive selection falls from 4.6% (0–6 kPa) to 2.6% and 1.8% (6–12, 12–18 kPa) in the uniform scenario. A stiffer cortex raised positive selection to 7.9% at 0–6 kPa, but the advantage vanished at higher stiffness (Figure-2).

Contact opportunity matters. Neglected cells had lower mean contact fractions (fraction of time steps spent in contact with a stromal cell) than selected cells. At the mid-range stiffness test, stromal density was the largest driver (negative selection 17%→56%); at the low-stiffness baseline this effect fell below noise. Similarly, influence of other parameters also depends on the stiffness range.

The contribution of mechanobiological amplification parameters was undetectable in the tested stiffness ranges. However, not detected is not the same as absent: priming equilibrates in minutes against a multi-day residence of that thymocyte, and the catch-bond effect is capped at ≈6% of k_{off} with current parameters.

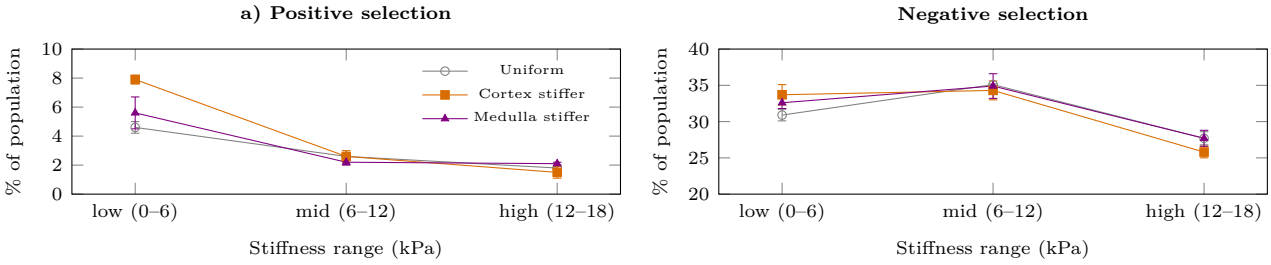


Figure-2: Selection outcome v/s stiffness ranges. Mean ± SD of three replicate runs per condition (1,000 cells each).

Positive selection	0–6	6–12	12–18
Uniform	4.6%	2.6%	1.8%
Cortex stiffer	7.9%	2.6%	1.5%
Medulla stiffer	5.6%	2.2%	2.1%

Negative selection	0–6	6–12	12–18
Uniform	30.9%	35.1%	27.7%
Cortex stiffer	33.7%	34.3%	25.8%
Medulla stiffer	32.6%	34.9%	27.7%

Main limitations

- Appropriate data sets do not currently exist for important parameters such as pre-selection thymocyte TCR-pMHC K_{off} values.
- Cytoskeletal priming and mechanosensitive competence act as a quasi-steady-state variables in current parameterization.
- Important physiological features (stromal cell, MHC-I/II and CD4/CD8/T_{reg} distinction, and dynamic nature of microenvironment) are excluded for simplicity.

Future work

- Recalibrate the k_{off} and signaling protein levels using more suited datasets.
- Add a slow (hours-scale) mechanosensitive state and test history dependence.
- Measure the thymocyte migration–stiffness curve to constrain K_{high}.
- Addition of physiological features that play important roles in the thymic selection process

Scoping Review: Mapping the literature of mechanobiology of thymic selection

Central question

What is known about the influence of stromal mechanical properties of the thymic stroma on thymic selection outcomes, and how aging and disease affect this relation?

Relevant Background

- **Stiffness data are sparse and the reported values are currently unresolved.**
- **Thymocyte mechanosensing is mostly inferred from peripheral T cells:** no thymocyte stiffness–migration or stiffness–signalling curves found confirm.

Search strategy

- PubMed and Scopus; dates from 01/01/2010 to 11/07/2026;
- Two complementary searches were devised to maximize retrieval of relevant literature for prioritization of sensitivity over specificity. For this purpose, separate search modules (SMs) were designed to identify studies addressing:
  - **SM1:** Thymic and thymocyte biology;
  - **SM2:** Mechanobiology and relevant mechanotransduction pathways;
  - **SM3:** Bioengineered thymic systems.
- These SMs were combined in the following manner:
  - **Search 1:** SM1 + SM2 (aimed to retrieve papers covering mechanotransduction pathways associated to thymic biomechanics and thymocyte biology);
  - **Search 2:** SM1 + SM3 (aimed to retrieve papers on bioengineered systems that focus on thymic biomechanics).
- 6,345 items obtained, 20 included after four screening stages (right).

Main findings:

- No study quantitatively linked measured thymic stiffness to selection outcomes. However, multiple studies discussed mechanisms in relevant contexts (such as catch-bond kinetics, TCR mechanotransduction) exploring which can form the missing link.
- No study discussed how thymic stromal mechanical properties change in ageing and disease and how this might influence selection outcomes.

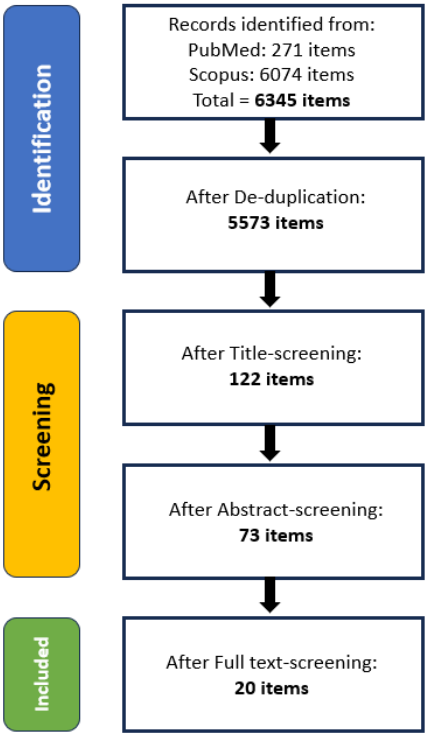


Figure-3: Study search and screening flow

Hypotheses generated from the model and review

Model Component Tests:

H1 — Gradient orientation is conditional on absolute stiffness

A stiffer cortex favours positive selection only when stiffness lies near the migration optimum, and the effect disappears above it. *Test:* cultures at matched mean stiffness with opposite internal gradients.

H2 — Contact opportunity influences neglect

Neglect outcome partly reflects reduced stromal contact, not only low TCR affinity. *Test:* relate stromal contact frequency and duration in thymic slices to later activation markers (e.g. CD69).

Model Mechanism Tests:

H3 — Migration optimum (supported by the model; assumption untested)

Thymocyte migration speed is biphasic in substrate stiffness. *Test:* track DP thymocytes on 0.5–30 kPa hydrogels; a monotonic or flat curve might falsify this assumption.

H4 — A slow mechanosensitive process is needed

Mechanical history can influence fate only through a process with a timescale of hours or longer. *Test:* stiffness-dependent ZAP70 phosphorylation and YAP/TAZ localisation in primary thymocytes over 30 min to 24 h.

Application Tests:

H5 — Ageing-related stiffening reduces selection through migration

If stiffening carries thymocytes beyond the migration optimum, it would add to the effect of epithelial loss. *Test:* compare migration speed and stromal contact in young and aged thymic slices with matched stromal density and AFM stiffness.